TEJAL GOYAL

604-782-2249 | itejalgoyal@gmail.com | linkedin.com/in/tejalgoyal | github.com/tejalgoyal2 | tgoyal.me

SUMMARY

Applied Data Science MEng and Software Engineer Co-op building and testing backend services, SQL-backed data layers, and APIs. Work in Python, C#, and Rust, with data structures and algorithms, version control, and CI/CD practice. Shipped a service on the Anthropic API that auto-resolved 50 of 83 records.

EDUCATION

MEng Applied Data Science *Jan 2025 – Dec 2026*
University of Victoria, BC *GPA: 8.82/9.0*
Relevant Coursework: Algorithms and Data Models; Systems for Massive Datasets; Artificial Intelligence; Data Analysis and Pattern Recognition

BEng Electronics and Computer Engineering *2020 – 2024*
Thapar Institute of Engineering & Technology, Patiala, India *GPA: 8.9/10.0*
Relevant Coursework: Data Structures and Algorithms; Object-Oriented Programming; Database Management Systems; Operating Systems

WORK EXPERIENCE

Software Engineer Co-op *May 2026 – Dec 2026*
British Columbia Investment Management Corporation (BCI), Victoria, BC

- Resolved data-layer defects in a .NET 8 financial ledger application, carrying tickets end to end from reproduction through code review to release
- Shipped a vendor onboarding service on the Anthropic API that resolves counterparty codes against an existing hierarchy with a confidence score and rationale per record, eliminating 3+ hours of weekly manual matching and auto-resolving 50 of 83 records
- Traced a report-to-source data discrepancy through a migrated stack of cloud storage, a semantic model, and layered transformations, isolating records dropped for a missing identifier and writing a two-key matching function that recovered them
- Build an AI-assisted Playwright regression suite for an Appian application, collapsing 40+ logged hours of manual click-through testing per cycle into a few clicks, with live-watch runs and a report of screenshots and recordings
- Replaced a manual quarterly routine of running database queries, verifying feed completeness, and chasing support on gaps with a Power Automate flow that validates 12 feed-months and branches to the finished report or an itemized alert
- Contribute to a flagship financial system rebuilt from a vendor-locked .NET/WPF stack into a React application on Azure with GitHub Copilot, removing licensing risk and enabling cloud-native deployment
- Coordinate user stories in a Scrum/Kanban hybrid with product owners, bringing Jira tickets into sprint planning and maintaining Confluence documentation across standups and backlog refinement

Cybersecurity and Risk Co-op *Jan 2026 – Apr 2026*
British Columbia Investment Management Corporation (BCI), Victoria, BC

- Developed a PowerShell tool authenticating against a REST API, walking a page tree, and parsing HTML into structured exports, cutting a manual data extraction from 3 hours to 10 minutes

Research Intern, Federated Learning *Jan 2024 – Jul 2024*
Indian Institute of Technology Ropar, Punjab, India

- Engineered a modular federated learning framework in PyTorch with configurable client fraction, batch size, and local epochs, training ResNet and VGG across 4 datasets at scales up to 1M simulated clients
- Built a utility to programmatically convert TensorFlow code to PyTorch, cutting cross-framework integration from 2 days of manual rewriting to hours

PROJECTS

Dime: AI Expense Tracker *May 2026*
Next.js 15, React 19, TypeScript, Cloudflare Workers, Supabase, Google Gemini | *dime.tgoyal.me*

- Built and deployed an AI expense tracker on Next.js 15, React 19, and Supabase, integrating Google Gemini for natural-language expense parsing and real-time categorization
- Implemented auth with Supabase SSR cookie management, per-route verification on every API endpoint, and Row Level Security enforcing isolation at the PostgreSQL level

Claubi: AI Coding Agent *Apr 2026*
Rust, Tokio, Ollama, Serde, async-trait | *github.com/tejalgoyal2/claubi*

- Architected a local-first AI coding agent in Rust with a Planner-Executor-Reviewer pipeline, routing tasks to local models via Ollama with NDJSON streaming over async mpsc channels

TECHNICAL SKILLS & CERTIFICATIONS

Languages & Fundamentals: Python, C#, Rust, SQL, JavaScript/TypeScript, Object-Oriented Programming, Data Structures and Algorithms
Backend & APIs: .NET 8, Node.js, REST API Design, Service-Oriented Architecture, System Design, SQL Server, PostgreSQL, MongoDB
Testing & DevOps: Unit Testing, Playwright, Git, GitHub, CI/CD, Monitoring, Docker, Kubernetes, Azure DevOps, Code Review, Linux
Cloud & AI: Microsoft Azure, Azure Data Factory, Databricks, Cloudflare Workers, Anthropic API, MCP (Model Context Protocol)